

Assignment 6 – Coding A Device Driver

Description:

The goal of the assignment is to write a device driver. The device driver should have functions of: read, write, open, release and ioctl function. The project also should not be trivial.

Approach:

Since I speak 2 languages, I thought it would be a great idea to write a Russian - English translator. My idea was to utilize the ioctl command by changing the mode between Russian to English and English to Russian translations. First, I need to write the dictionary of words, and their translations and store it. I was planning to be able to translate sentences, not only words. But for that, first I need to translate each word. So I have the function translate word, which will check what mode was the request made in, go through the dictionary and return the translated word into user's buffer, and return the word not translated if it is not the dictionary.

To be able to translate the whole sentence, I have a translate_sentence function. The function goes over the input buffer, taking the words one by one, sending them to the translate_word function, getting the translation and putting it in the input buffer.

To use the translator, first we need to run the program from the module, to run the driver. We also need to install the driver with: sudo insmod translator.ko. Then we need to run it on test. Once it runs, we need to choose what way to translate, 0 for Ru->En and 1 for En->Ru. Then include the sentence we need to translate, and it will output the translation.

Issues and Resolutions:

My first issue was that whenever I tried to run the test with make run, it gave me a permission denied error. So I had to run it with sudo.

Another issue I had is that I needed to remove and install the driver again, whenever I made some changes to the dictionary.

Also, some of the words do not translate directly from english to russian and vice versa, like word are or is in english have no translation. So i had to leave it empty, which created double space in output buffer. Also, the translation is not 100% accurate, since I could not include phrases.

```
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ make run
./Almat_Dias_HW6_main
Failed to open the device: Permission denied
make: *** [Makefile:59: run] Error 1
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ sudo make run
```

Analysis:

Screen shot of compilation:

```
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Module$ make
make -C /lib/modules/`uname -r`/build M=/home/student/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Module modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-7-generic'
make[2]: Entering directory '/home/student/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Module'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: aarch64-linux-gnu-gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0
You are using: gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0
CC [M] translator.o
MODPOST Module.symvers
CC [M] translator.mod.o
CC [M] .module-common.o
LD [M] translator.ko
BTF [M] translator.ko
make[2]: Leaving directory '/home/student/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Module'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-7-generic'
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Module$ █

student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Module$ sudo insmod translator.ko
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Module$ █

student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ make
gcc -c -o Almat_Dias_HW6_main.o Almat_Dias_HW6_main.c -g -I.
gcc -o Almat_Dias_HW6_main Almat_Dias_HW6_main.o -g -I. -l pthread
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ █
```

Screen shot(s) of the execution of the program:

```
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ sudo make run
./Almat_Dias_HW6_main
Enter 0 for RU->EN or 1 for EN->RU: 1
Enter a sentence to translate: hello how are you I love you
Translated sentence: привет как ты я любовь ты
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ █

student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ sudo make run
./Almat_Dias_HW6_main
Enter 0 for RU->EN or 1 for EN->RU: 0
Enter a sentence to translate: привет как ты я люблю тебя
Translated sentence: hello how you I love you
student@student:~/CSC415/assignment6/csc415-assignment-6-device-driver-vincivv/Test$ █
```